

§11.2 - Series

Definition #1 (Infinite Series)

Let (a_n) be a sequence.

Then

$$\sum_{k=1}^{\infty} a_k = a_1 + a_2 + a_3 + \cdots$$

is an **infinite series**.

We have

$$\sum_{k=1}^{\infty} a_k = \sum_{n=1}^{\infty} a_n,$$

or the index of our sum is a "dummy" variable.

However

$$\sum_{k=1}^{\infty} a_k \neq \sum_{n=1}^{\infty} a_k$$

since

$$\sum_{n=1}^{\infty} a_k = a_k + a_k + a_k + \dots$$

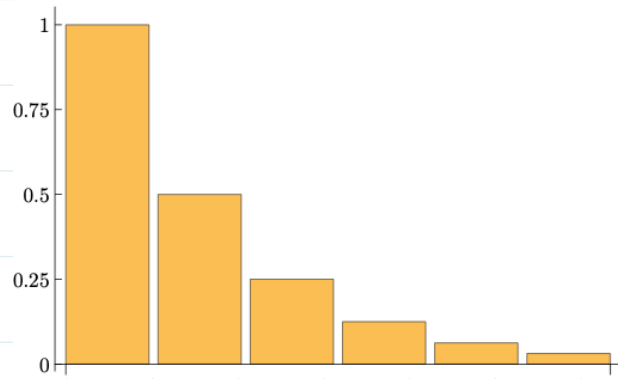
while

$$\sum_{k=1}^{\infty} a_k = a_1 + a_2 + a_3 + \dots$$

We can think of

$$\sum_{n=1}^{\infty} a_n$$

as integrating the sequence (a_n) over $[1, \infty)$.



Definition #2 (Partial Sum)

Let (a_n) be a sequence.

Then

$$S_n = \sum_{k=1}^n a_k = a_1 + a_2 + \cdots + a_n$$

is the **n th partial sum** of the series.

Let

$$S_n = \sum_{k=1}^n k = 1 + 2 + 3 + \dots + n.$$

Then

$$S_1 = 1$$

$$S_2 = 1 + 2 = 3$$

$$S_3 = 1 + 2 + 3 = 6.$$

One can show

$$S_n = \frac{1}{2}n(n+1)$$

and so

$$S_n \rightarrow \infty$$

as $n \rightarrow \infty$.

If

$$S_n = \sum_{k=1}^n a_k$$

then

$$\begin{aligned} S_{n+1} &= \sum_{k=1}^{n+1} a_k \\ &= \underbrace{a_1 + \dots + a_n}_{S_n} + a_{n+1} \\ &= S_n + a_{n+1}. \end{aligned}$$

Definition #3 (Convergence)

If

$$\lim_{n \rightarrow \infty} S_n = S \in \mathbf{R}$$

then the series $\sum a_n$ **converges**.

If (S_n) diverges then the series $\sum a_n$ **diverges**.

We are requiring the partial sums to converge,

$$\lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} \sum_{k=1}^n a_k = \sum_{k=1}^{\infty} a_k.$$

If we have

$$S_n = a_1 + \dots + a_n = \frac{5n+1}{6n-7}$$

then

$$S_n \rightarrow \frac{5}{6}$$

implies

$$\sum_{k=1}^{\infty} a_k = \frac{5}{6}.$$

Remark

Here

1. $\sum a_n$ or $\sum a_k$: denotes the series.

2. $\sum_{n=1}^{\infty} a_n$: is the value of the series starting at $n = 1$.

3. $\sum_{n=2}^{\infty} a_n$: is the value of the series starting at $n = 2$.

An analog for functions is the following:

$$f : \mathbb{R} \rightarrow \mathbb{R}$$

where

$$f(x) = x^2.$$

Then

- 1. f is the name of the function.**
- 2. $f(x)$ is the value of the function at x .**
(In mathematics, using $f(x)$ when referring to the function f is common practice.)

Lemma #1

Let $r \in \mathbf{R}$.

Then

$$\lim_{n \rightarrow \infty} r^n = \begin{cases} 1, & r = 1 \\ 0, & r \in (-1, 1). \end{cases}$$

For any other r , the sequence diverges.

Here

$$\lim_{n \rightarrow \infty} 1^n = 1$$

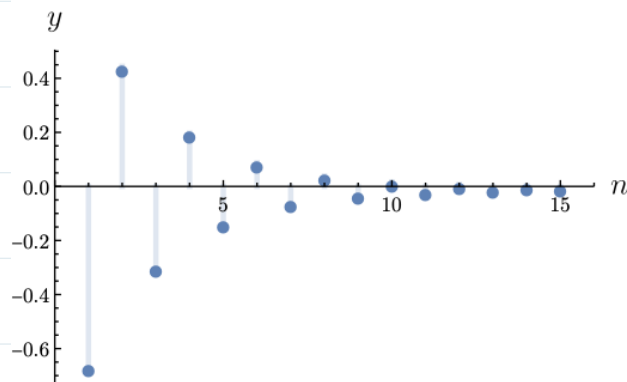
while

$$\lim_{n \rightarrow \infty} (-1)^n \text{ diverges (d.n.e).}$$

If $|r| < 1$ then

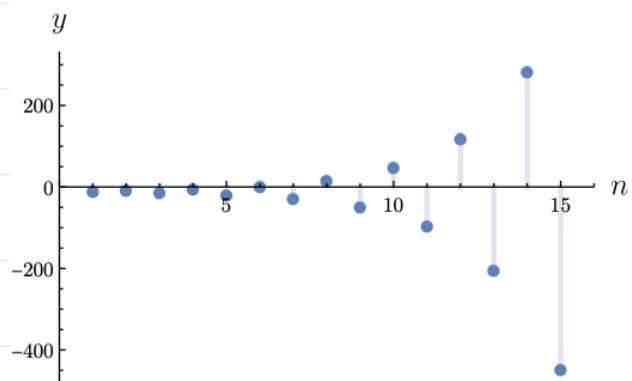
$$\lim_{n \rightarrow \infty} r^n = 0$$

since we have exponential decay.



While $|r| > 1$ yields

$$\lim_{n \rightarrow \infty} |r|^n = \infty.$$



Definition #4 (Geometric Series)

Let $r \in \mathbf{R}$.

The **geometric series**

$$\begin{aligned}\sum_{k=0}^{\infty} r^k &= 1 + r + r^2 + \dots \\ &= \frac{1}{1 - r}\end{aligned}$$

exists only for $|r| < 1$.

Let

$$S_n(r) = \sum_{k=0}^n r^k = 1 + r + r^2 + \dots + r^n$$

then

$$\begin{aligned} S_n(r) - rS_n(r) &= 1 + r + r^2 + \dots + r^n \\ &\quad - (r + r^2 + \dots + r^{n+1}) \\ &= 1 - r^{n+1}. \end{aligned}$$

We then factor the lefthand side, and get

$$(1 - r)S_n(r) = 1 - r^{n+1}$$

$$S_n(r) = \frac{1 - r^{n+1}}{1 - r}$$

for $r \neq 1$.

If $|r| < 1$ then $|r^{n+1}| \rightarrow 0$ as $n \rightarrow \infty$ and so

$$\frac{1}{1 - r} = \lim_{n \rightarrow \infty} S_n(r) = \sum_{k=0}^{\infty} r^k.$$

Remark

Observe,

$$\sum_{k=0}^{\infty} 0^k = 1 + 0 + 0 + \dots$$

means we are defining $0^0 = 1$ in our summation.

Corollary #1

We have

$$\sum_{k=1}^{\infty} r^k = \frac{r}{1-r}$$

for $|r| < 1$.

We can remove the constant term in the Geometric series,

$$\sum_{k=0}^{\infty} r^k = 1 + r + r^2 + \dots$$

$$\sum_{k=1}^{\infty} r^k = r + r^2 + \dots$$

gives

$$\begin{aligned}\sum_{k=1}^{\infty} r^k &= \left(\sum_{k=0}^{\infty} r^k \right) - 1 \\ &= \frac{1}{1-r} - 1 \\ &= \frac{1}{1-r} - \frac{1-r}{1-r} \\ &= \frac{r}{1-r}.\end{aligned}$$

Example #1

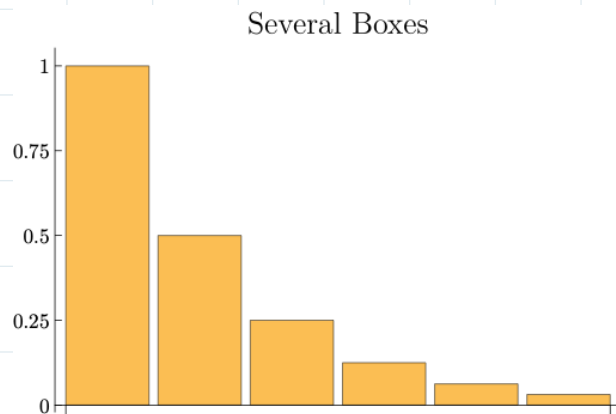
Find

$$\sum_{k=0}^{\infty} 2^{-k}.$$

We have a Geometric sum with $r = 1/2$:

$$\sum_{k=0}^{\infty} \left(\frac{1}{2}\right)^k = \frac{1}{1-1/2} = 2.$$

The sum corresponds to the area of a sequence of boxes where each box is half the height of the last and the initial box has height 1.



Example #2

Find

$$\sum_{k=1}^{\infty} \frac{2}{(2k+1)(2k+3)}.$$

Here

$$S_n = \sum_{k=1}^n \frac{2}{(2k+1)(2k+3)}$$

Just like with integration of rational functions, apply a PFD:

$$\frac{2}{(2k+1)(2k+3)} = \frac{A}{2k+1} + \frac{B}{2k+3}$$
$$2 = A(2k+3) + B(2k+1)$$

If $k = -\frac{1}{2}$,

$$2 = A(-1+3) \Rightarrow A = 1.$$

If $k = -\frac{3}{2}$

$$2 = B(-3+1) \Rightarrow B = -1.$$

Then

$$S_n = \sum_{k=1}^n \left(\frac{1}{2k+1} - \frac{1}{2k+3} \right)$$
$$= \left(\frac{1}{3} - \frac{1}{5} \right) + \left(\frac{1}{5} - \frac{1}{7} \right) + \left(\frac{1}{7} - \frac{1}{9} \right) + \dots$$
$$+ \left(\frac{1}{2n-1} - \frac{1}{2n+1} \right) + \left(\frac{1}{2n+1} - \frac{1}{2n+3} \right)$$

and notice all the terms that cancel! This

simplifies to

$$S_n = \frac{1}{3} - \frac{1}{2n+3}.$$

We call this a telescoping sum.

Now

$$S_n \rightarrow \frac{1}{3}$$

since

$$\frac{1}{2n+3} \rightarrow 0$$

as $n \rightarrow \infty$. This shows

$$\sum_{k=1}^{\infty} \frac{2}{(2k+1)(2k+3)} = \frac{1}{3}.$$

Theorem #1

If

$$\sum a_n$$

converges, then

$$a_n \rightarrow 0.$$

If

$$\sum_{k=1}^{\infty} a_k = S \in \mathbb{R}$$

then

$$S_n \rightarrow S \text{ and } S_{n+1} \rightarrow S$$

and

$$S_{n+1} - S_n = \sum_{k=1}^{n+1} a_k - \sum_{k=1}^n a_k = a_{n+1}.$$

Thus

$$S_{n+1} - S_n \rightarrow S - S = 0 \Rightarrow a_{n+1} \rightarrow 0.$$

This is a consistency condition, all convergent sums $\sum a_n$ must satisfy

$$a_n \rightarrow 0.$$

Corollary #2 (Basic Divergence Test)

Let (a_n) be a sequence.

If

$$\lim_{n \rightarrow \infty} a_n \neq 0 \quad \text{or} \quad \lim_{n \rightarrow \infty} a_n \text{ d.n.e.}$$

then

$$\sum a_n$$

diverges.

The B.D.T. gives an immediate argument
to why

$$\sum r^n$$

diverges whenever $|r| \geq 1$:

$$\lim_{n \rightarrow \infty} r^n \text{ d.n.e.}$$

for $|r| \geq 1$.

Example #3

Determine if

$$\sum n \sin \left(\frac{1}{n} \right)$$

converges.

Here

$$\lim_{n \rightarrow \infty} n \sin\left(\frac{1}{n}\right) = \lim_{h \rightarrow 0^+} \frac{\sin(h)}{h} = 1$$

by geometry or l'Hospital. Thus

$$\sum n \sin\left(\frac{1}{n}\right)$$

diverges by the B.D.T.

Warning

There exists a sequence (a_n) such that

$$a_n \rightarrow 0 \quad \text{and} \quad \sum a_n \text{ diverges.}$$

We will show this explicitly in §11.3.

Theorem #2

If $\sum a_n$ and $\sum b_n$ exist then

$$\sum_{k=1}^{\infty} (c \cdot a_k + b_k) = c \sum_{k=1}^{\infty} a_k + \sum_{k=1}^{\infty} b_k.$$

for any $c \in \mathbf{R}$.

Example #4

Express the numbers

$$0.1111111111\dots \quad \text{and} \quad 0.1212121212\dots$$

as ratios of integers.

Notice that

$$0.\text{IIIIII}\dots = \frac{1}{10} + \frac{1}{100} + \frac{1}{1,000} + \dots$$

$$= \sum_{k=1}^{\infty} \left(\frac{1}{10}\right)^k$$

$$= \frac{\frac{1}{10}}{1 - \frac{1}{10}}$$

$$= \frac{\frac{1}{10}}{\frac{9}{10}}$$

$$= \frac{1}{9}$$

and

$$\begin{aligned}
0.010101\dots &= \frac{1}{100} + \frac{1}{10,000} + \frac{1}{1,000,000} + \dots \\
&= \frac{1}{100} + \frac{1}{100^2} + \frac{1}{100^3} + \dots \\
&= \sum_{k=1}^{\infty} \left(\frac{1}{100}\right)^k \\
&= \frac{\frac{1}{100}}{1 - \frac{1}{100}} \\
&= \frac{\frac{1}{100}}{\frac{99}{100}} \\
&= \frac{1}{99}.
\end{aligned}$$

Hence

$$\begin{aligned}
0.121212\dots &= 0.111111\dots + 0.010101\dots \\
&= \frac{1}{9} + \frac{1}{99} \\
&= \frac{11}{99} + \frac{1}{99} \\
&= \frac{12}{99} \\
&= \frac{4}{33}.
\end{aligned}$$

Example #5

Find the values of x such that

$$\sum_{k=0}^{\infty} e^{kx}$$

converges.

For $k \in \mathbb{Z}$

$$e^{kx} = (e^x)^k$$

and $|e^x| < 1$ whenever $x < 0$.

We have a convergent Geometric sum

$$\sum_{k=0}^{\infty} e^{kx} = \sum_{k=0}^{\infty} (e^x)^k = \frac{1}{1 - e^x}$$

given $x < 0$. If $x \geq 0$, the sum diverges.